

Integer part and fractional part functions



- **Author:** Fabio Furini.
- **Book:** “Analisi matematica 1”, Marco Bramanti, Carlo D. Pagani and Sandro Salsa, Zanichelli.

Contents

1	Integer part and fractional part functions	2
2	Piecewise-defined functions	4

1 Integer part and fractional part functions

- Two functions that are typically encountered when writing *algorithms* are the integer part (floor) function and the fractional part (or decimal part) function.

Definition 1: of the integer part function

The **integer part (floor) function** is:

$$f : \mathbb{R} \rightarrow \mathbb{Z}, x \mapsto \lfloor x \rfloor \quad (\text{or } x \mapsto \lfloor x \rfloor) \quad (1)$$

Definition 2: of the upper integer part function

The **upper integer part (ceiling) function** is:

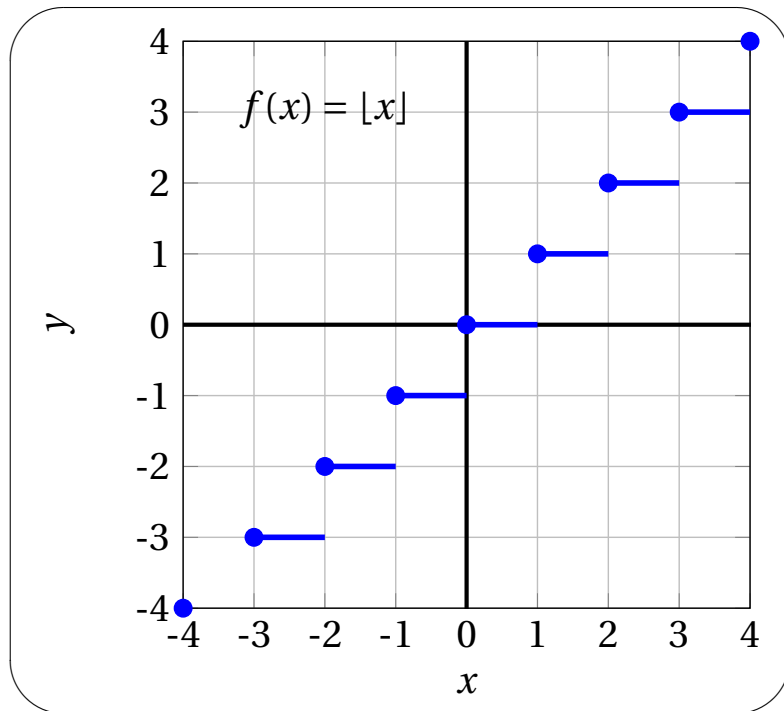
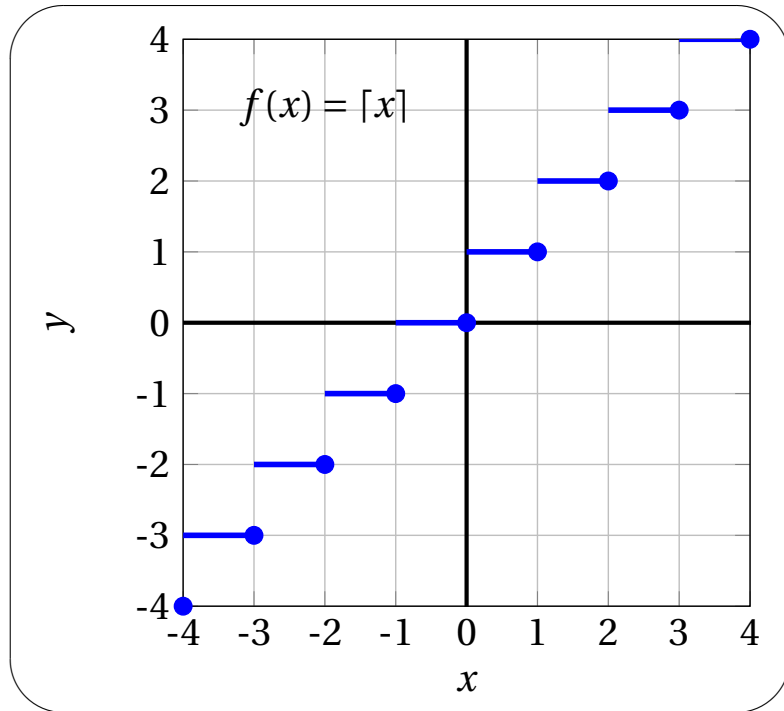
$$f : \mathbb{R} \rightarrow \mathbb{Z}, x \mapsto \lceil x \rceil \quad (2)$$

Definition 3: of the fractional part function

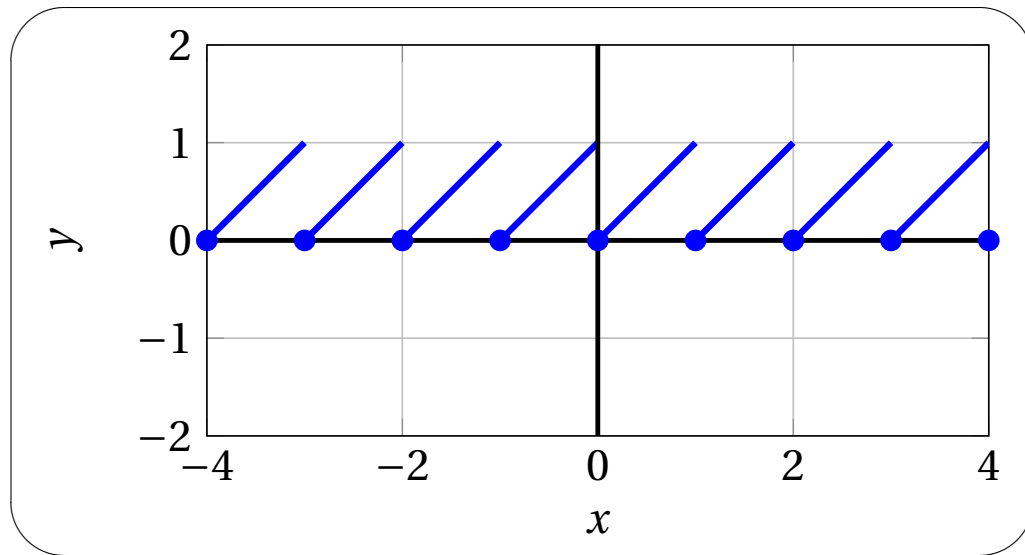
The **fractional part function** is:

$$f : \mathbb{R} \rightarrow [0, 1), x \mapsto \{x\} \quad (3)$$

- The integer part function $f(x) = \lfloor x \rfloor$ is *monotone increasing*, like the upper integer part function $f(x) = \lceil x \rceil$.
- Graphs of the functions $f(x) = \lceil x \rceil$ and $f(x) = \lfloor x \rfloor$ on the interval $[-4, 4]$.



- Note that the fractional part is a periodic function with period 1:



2 Piecewise-defined functions

- Starting from the elementary functions, new ones can be built by using **different analytic definitions on different intervals**.

Definition 4: of piecewise-defined function

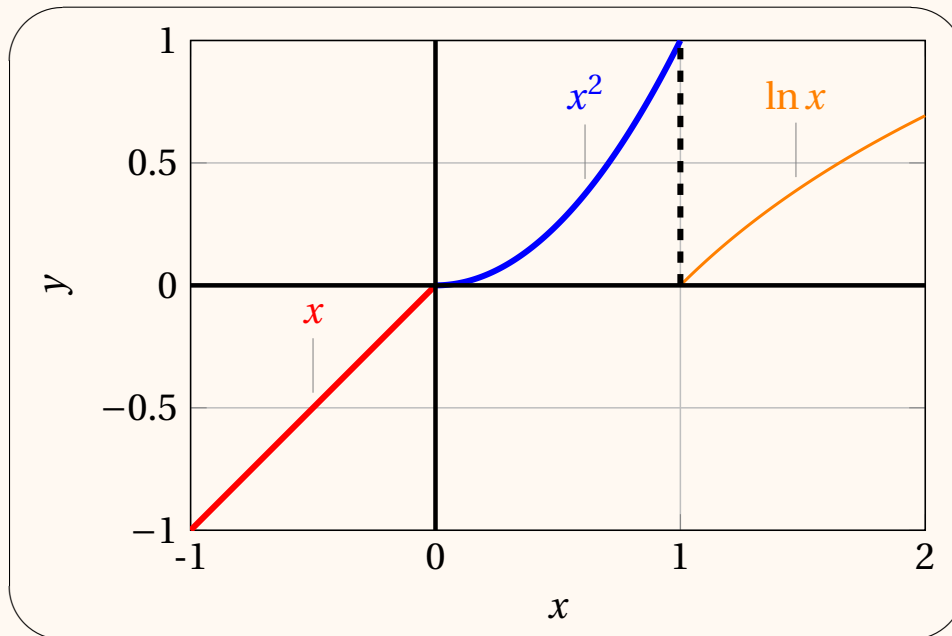
A function f whose value $f(x)$ is computed through different “instructions” depending on the interval in which x lies is called a **piecewise-defined function**.

Example 1: Piecewise-defined functions

Consider for example

$$f(x) = \begin{cases} \ln x & \text{if } x > 1, \\ x^2 & \text{if } 0 < x \leq 1, \\ x & \text{if } x \leq 0 \end{cases}$$

its graph is:



- From the *logical / algorithmic* point of view, one can say that a function of this kind has the peculiarity of being built using not only mathematical functions but also the **logical function** “if ... then”.